

Nonlinear Dynamics & Chaos HW2

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Technion

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1 Quadratic Map

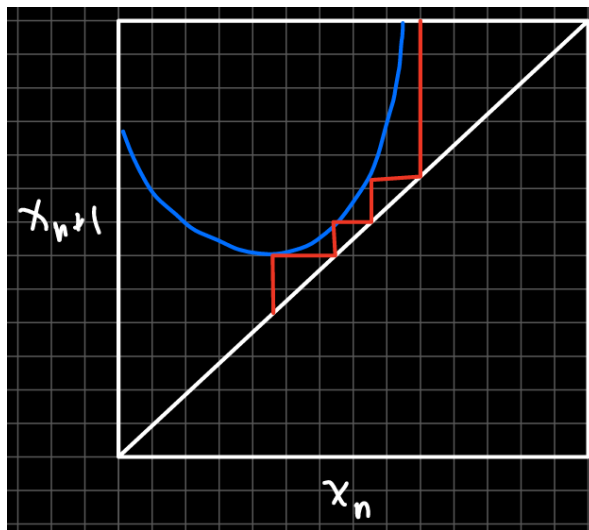
Consider the one-parameter family of maps

$$x_{n+1} = x_n^2 + c, \quad c \in \mathbb{R} \quad (1)$$

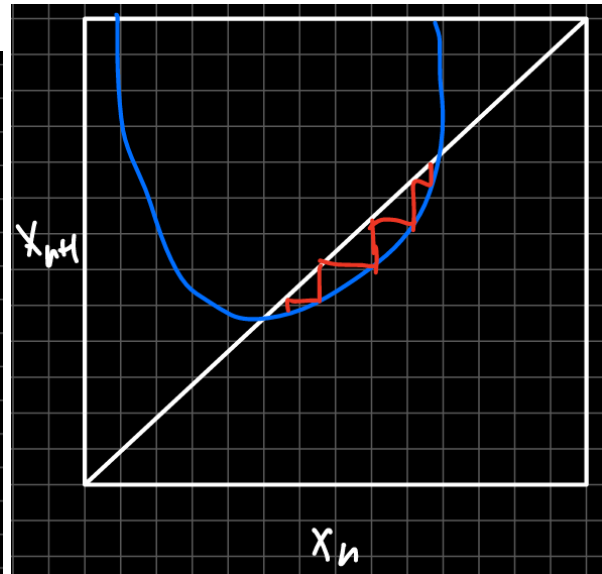
1. Draw cobweb plots of the map for $c = 0.4$ and $c = 0$. Briefly describe the qualitative behavior you read from each cobweb
2. Find all fixed points as a function of c and classify their linear stability
3. Find the values of c at which bifurcations occur and write the type of bifurcation for each value
4. You should have found a value of c at which a fixed point loses stability but no new fixed point is born. Show that instead a period-2 orbit emerges. find p and q explicitly as functions of c .
5. Find the range of c for which the 2-cycle is stable.
6. Sketch a partial bifurcation diagram in the c, x plane for $c \in [-1.5, 0.5]$. Mark the fixed point branches and the 2-cycle branch, and mark the bifurcation values from (c).

1.1 cobwebs

The function we'll use to draw is $f(x) = x^2 + c$. On the left cobweb, we can see that the behavior is unbound



(a) Cobweb diagram for $c = 0.4$



(b) Cobweb diagram for $c = 0$

and will tend to infinity. On the right cobweb, $x = 1$ is a source and $x = 0$ is a stable sink. points beyond $x = 1$ will go to infinity. points before $x = 0$ will converge to $x = 0$.

1.2 Fixed points and Stability

To find the fixed points we solve

$$x^2 - x + c = 0 \quad (2)$$

whose solutions are

$$x_{\pm} = \frac{1 \pm \sqrt{1-4c}}{2} \quad (3)$$

looking at the derivative at those points and restricting to only real values

$$f'(x_{\pm}) = 1 \pm \sqrt{1-4c} = \begin{cases} f'(x_+) = 1 + \sqrt{1-4c} \geq 1 \\ f'(x_-) = 1 - \sqrt{1-4c} \end{cases} \quad (4)$$

to check the second point's stability we require that

$$-1 < 1 - \sqrt{1-4c} < 1 \quad (5)$$

which is satisfied for

$$-\frac{3}{4} < c < \frac{1}{4} \quad (6)$$

in summary, x_+ is always unstable, and x_- is stable for $-\frac{3}{4} < c < \frac{1}{4}$.

1.3 Bifurcations and Types

Using the last section we see two points:

1. $c = \frac{1}{4}$. Both fixed points combine and disappear for $c > \frac{1}{4}$. When that happens the derivative is exactly +1 so the graph of f is tangent to the diagonal, meaning this is a tangent bifurcation.
2. $c = -\frac{3}{4}$. No new fixed point is created so this is a period-doubling bifurcation.

1.4 period-2 orbit

period-2 points are where $f^2(x) = x$ so in our case

$$0 = (x^2 + c)^2 + c - x = x^4 + 2cx^2 - x + c^2 + c = 0 \quad (7)$$

but since we have two fixed points from the original $f(x)$ we can treat them as solutions and factor them out using polynomial division giving

$$f^2(x) - x = (x^2 - x + c)(x^2 + x + c + 1) \quad (8)$$

which means the 2-cycle is the roots of the second element

$$p, q = \frac{-1 \pm \sqrt{-4c-3}}{2} \quad (9)$$

we can notice that for $c = -\frac{3}{4}$ both roots coincide meaning the 2-orbit emerges from the fixed point associated with that value of c .

1.5 Stability range

Taking the derivative

$$(f^2)'(p) = f'(f(p)) \cdot f'(p) = f'(q) \cdot f'(p) = 4pq \quad (10)$$

then using Vieta on the second element we have

$$pq = c + 1 \quad (11)$$

so the multiplier is

$$\mu = 4c + 4 \quad (12)$$

and to have stability

$$-1 < 4c + 4 < 1 \implies -\frac{5}{4} < c < -\frac{3}{4} \quad (13)$$

1.6 bifurcation diagram

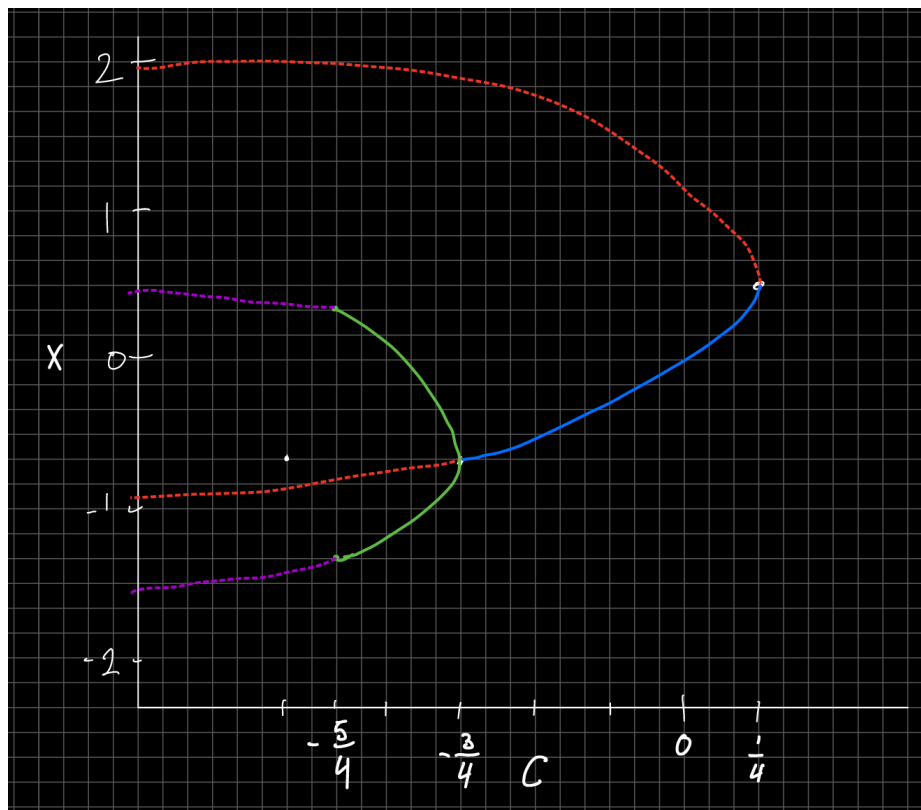


Figure 2 – bifurcation diagram. Solid lines are stable, dashed are unstable. The tangent bifurcation is seen at $c = 0.25$, and the flip at $c = -0.75$. At $c = -1.25$ the 2-cycle destabilises.

2 Newton Method as an iterated map

1. Write down the Newton map explicitly for $g(x) = x^2 - 4$. Show that it has exactly two fixed points that coincide with the roots of g , and that both are superstable in the sense of the definition above.
2. Starting from $x_0 = 1$ iterate and report the first few iterates $x_1, x_2, \dots$ until the result agrees with $x^* = 2$ to your tool's precision
3. Return to a general twice differentiable g and let x^* be a root, $g(x^*) = 0$. Show that x^* is a superstable fixed point of the newton map if $g'(x^*) \neq 0$. What happens if $g'(x^*) = 0$?

2.1 Calibration

Since

$$g'(x) = 2x \quad (14)$$

we have

$$f(x) = x - \frac{x^2 - 4}{2x} = \frac{x^2 + 4}{2x} = \frac{x}{2} + \frac{2}{x} \quad (15)$$

to find the fixed points we look at

$$f(x) = x \implies \frac{x}{2} = \frac{2}{x} \implies x = \pm 2 \quad (16)$$

the multiplier for these points is

$$f'(x) = \frac{1}{2} - \frac{2}{x^2} \implies f'(\pm 2) = \frac{1}{2} - \frac{1}{2} = 0 \quad (17)$$

meaning both of the fixed points are superstable.

2.2 Numerical Iteration

Our series is

$$x_{n+1} = \frac{x_n}{2} + \frac{2}{x_n} \quad (18)$$

so, using a simple python script At iteration number 6 the value is so close to 2 that a 64-bit floating point

n	x_n
0	1
1	2.5
2	2.05
3	2.000610
4	2.000000093
5	2.000000000000002
6	2

Table 1 – Iterates from $x_0 = 1$

number can no longer capture the difference.

2.3 Generalisation

For

$$f(x) = x - \frac{g(x)}{g'(x)} \quad (19)$$

the multiplier is

$$f'(x) = 1 - \frac{g'(x)^2 - g(x)g''(x)}{g(x)^2} = \frac{g(x)g''(x)}{g'(x)^2} \quad (20)$$

it's immediate that if

$$g(x^*) = 0 \qquad g'(x^*) \neq 0 \quad (21)$$

then

$$f'(x^*) = 0 \quad (22)$$

making x^* superstable. If instead $g'(x^*) = 0$ then the Newton map formula is undefined. Instead if the root has multiplicity $m > 1$ then we can remove the singularity and get $f'(x^*) = \frac{m-1}{m}$ making the root attracting but not superstable.

3 Analysis of 2D linear systems

Part 1: Direct analysis. For each of the following 2D linear systems with $\dot{\mathbf{x}} = A\mathbf{x}$, compute the eigenvalues and (when real) the eigenvectors of A , and sketch the phase portrait. Indicate the directions of the eigenvectors when real, the sense of rotation when complex, and a few representative trajectories.

1. $\dot{x} = x + 3y, \quad \dot{y} = 2x$

2. $\dot{x} = -2x - 4y, \quad \dot{y} = x - 2y$

Part 2: Classification from trace and determinant alone. For the matrix below, classify the fixed point at the origin using only $T = \text{tr } A$ and $\Delta = \det A$. No need to compute the eigenvalues explicitly.

$$A = \begin{pmatrix} -3 & 1 \\ 2 & -4 \end{pmatrix} \quad (23)$$

3.1 Direct analysis

3.1.1 first one

rewriting into matrix form

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 0 \end{pmatrix} \quad (24)$$

We find the eigenvalues

$$\det \begin{pmatrix} 1 - \lambda & 3 \\ 2 & -\lambda \end{pmatrix} = \lambda(\lambda - 1) - 6 = \lambda^2 - \lambda - 6 = (\lambda - 3)(\lambda + 2) = 0 \quad (25)$$

giving us the eigenvalues

$$\lambda = 3 \qquad \qquad \qquad \lambda = -2 \quad (26)$$

and to find the eigenvectors

$$\begin{pmatrix} -2 & 3 \\ 2 & -3 \end{pmatrix} v = 0 \implies v = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \quad (27)$$

for $\lambda = 3$, meaning trajectories move outwards along this vector and

$$\begin{pmatrix} 3 & 3 \\ 2 & 2 \end{pmatrix} v = 0 \implies v = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (28)$$

for $\lambda = -2$, meaning trajectories move inwards along this vector.

3.1.2 second one

Again

$$A = \begin{pmatrix} -2 & -4 \\ 1 & -2 \end{pmatrix} \quad (29)$$

with eigenvalues

$$\det \begin{pmatrix} -2 - \lambda & -4 \\ 1 & -2 - \lambda \end{pmatrix} = (2 + \lambda)^2 + 4 = \lambda^2 + 4\lambda + 8 = 0 \quad (30)$$

therefore

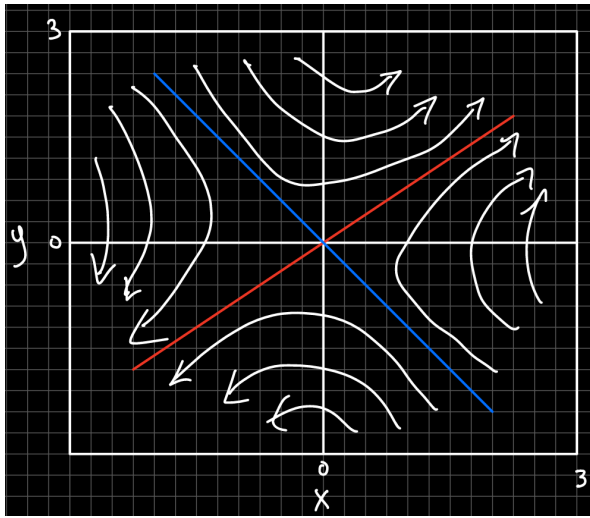
$$\lambda_{1,2} = \frac{-4 \pm \sqrt{16 - 32}}{2} = \frac{-4 \pm 4i}{2} = -2 \pm 2i \quad (31)$$

now if we plug in, for example, $x = 1, y = 0$ we get

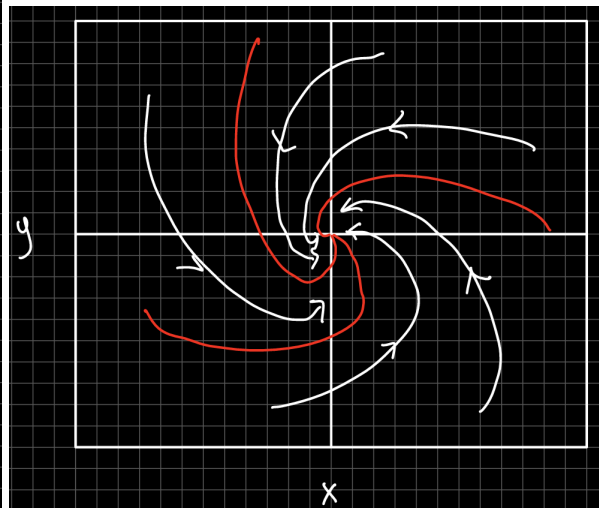
$$\dot{x} = -x \qquad \dot{y} = 1 \quad (32)$$

meaning this would move up and to the left, implying a counter clockwise rotation spiral inwards.

3.1.3 Figures



(a) Saddle, $\lambda = 3, -2$



(b) Stable spiral, $\lambda = -2 \pm 2i$

3.2 Classification

The trace and determinant are

$$\text{tr } A = -7 \qquad \det A = 10 \quad (33)$$

therefore the discriminant is

$$\text{tr}^2 - 4\Delta = 49 - 40 = 9 > 0 \quad (34)$$

meaning there are two real distinct eigenvalues. A positive determinant means they share a sign and the negative trace makes the sign negative. Therefore the origin is a stable node.

4 Transient Amplification by a non-normal matrix

Consider the two matrices

$$A_N = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix}, \quad A_{NN} = \begin{pmatrix} -1 & 5 \\ 0 & -2 \end{pmatrix} \quad (35)$$

1. Write the eigenvalues and eigenvectors of the two matrices. Are the eigenvectors of A_{NN} orthogonal to each other?
2. Sketch the phase portrait for the two cases. Discuss the difference between the two cases in how trajectories approach the origin at long times.
3. For the initial condition $\mathbf{x}_0 = (0, 1)$ write the solution $(x(t), y(t))$ explicitly for the two cases, sketch the trajectory on the phase plane.
4. Show that the norm of the solution $|\mathbf{x}(t)|$ for A_{NN} grows to be larger than the initial value before it decays, explain geometrically why that happens.

4.1 eigenthings

the first matrix is diagonal so the eigenvalues are $\lambda = -1, -2$ with eigenvectors $(1, 0)$ and $(0, 1)$. As for the 2nd matrix

$$\det \begin{pmatrix} -1 - \lambda & 5 \\ 0 & -2 - \lambda \end{pmatrix} = (1 + \lambda)(2 + \lambda) = 0 \quad (36)$$

giving eigenvalues $\lambda = -1, -2$. The eigenvectors

$$\begin{pmatrix} 0 & 5 \\ 0 & -1 \end{pmatrix} v = 0 \implies v = (1, 0) \quad (37)$$

and

$$\begin{pmatrix} 1 & 5 \\ 0 & 0 \end{pmatrix} v = 0 \implies v = (5, -1) \quad (38)$$

we can see the eigenvectors of A_{NN} are not orthogonal to each other.

4.2 Phase portrait

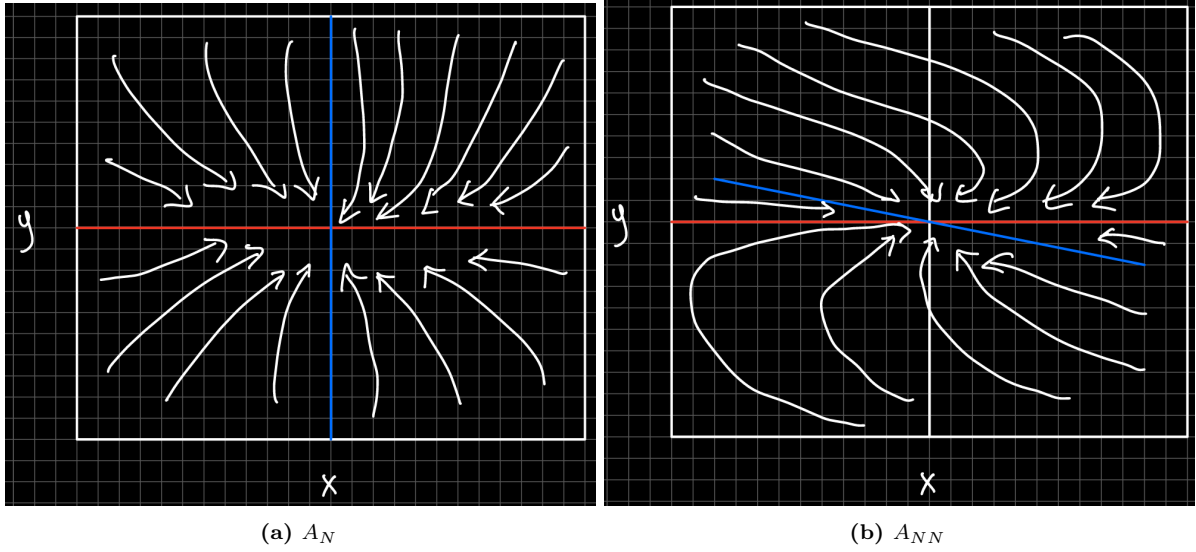
4.3 Explicit solution

Since A_N is diagonal each element is decoupled from the other giving us the solutions

$$x(t) = 0 \quad y(t) = e^{-2t} \quad (39)$$

for A_{NN} we can solve the equation for y immediately and obtain

$$y(t) = e^{-2t} \quad (40)$$



and use this solution for the equation for x

$$\dot{x} = -x + 5e^{-2t} \implies x(t) = Ce^{-t} - 5e^{-2t} \quad (41)$$

inserting $x(0) = 0$ we get $C = 5$ so

$$x(t) = 5(e^{-t} - e^{-2t}) \quad (42)$$

or in other notation

$$\mathbf{x}(t) = (5(e^{-t} - e^{-2t}), e^{-2t}) \quad (43)$$

4.4 growing decaying norm

Taking the norm of the solution

$$N^2 = \|\mathbf{x}(t)\|^2 = 25(e^{-t} - e^{-2t})^2 + e^{-4t} \quad (44)$$

redefining parameters to $u = e^{-t}$

$$N^2 = \|\mathbf{x}(t)\|^2 = 25(u - u^2)^2 + u^4 = 26u^4 - 50u^3 + 25u^2 \quad (45)$$

plugging into wolfram alpha to find the maximum gives us that there is a local maximum at

$$t^* \approx 0.648 \quad N \approx \sqrt{1.63} > 1 \quad (46)$$

with no other extrema for $t > t^*$. But

$$N(t=0) = 1 \quad (47)$$

therefore the norm grows to be larger than the initial value, but then decays.